f. DUCT SUPPORTS

Description - Support from building shaft

The Duct should be simply supported by Support Channel made up of cold rolled steel of quality **DX51 or greater and as per EC3(Euro code 3) or DIN EN 1993-1-1**

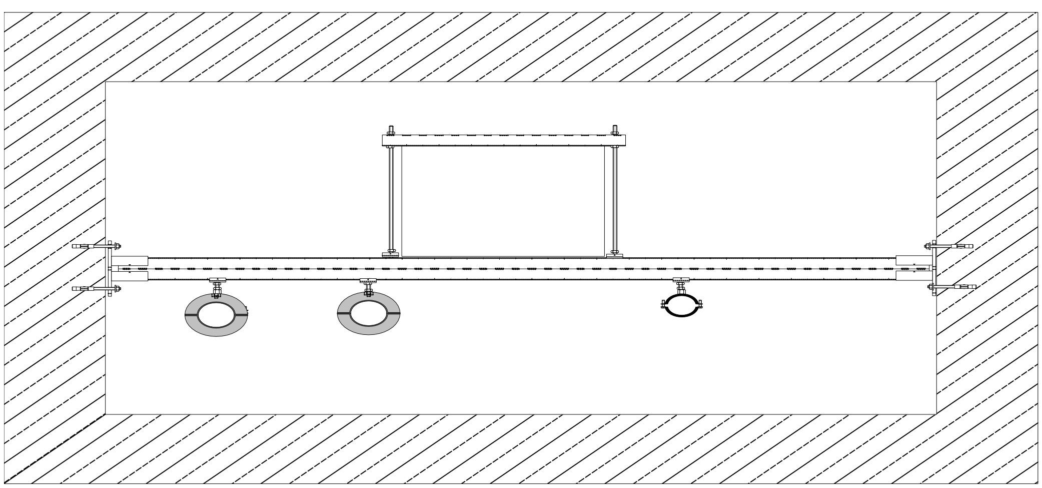
The Support channel should be **pre galvanized with minimum GSM of 275** and should have universal mounting slot on the front of the rail for accurate positioning of fasteners and system compatible **round and long holes on back of the rail**.

The Mounting according to static requirements should undertake into account the manufacturer's documents and should be monitored according to **RAL - GZ 655-C**.
The Threaded Rods used for the suspension of the Duct should be made up of **partially annealed** medium carbon steel of **grade 4.8 strength class and as per DIN 976 standard.** The Drop-in anchors or stud anchor used for the suspension of the rods should be **ETA (EUROPEAN TECHNICAL APPROVAL) with CE mark** for cracked and un-cracked concrete. It should be divided into four expansion segments for uniform pressing force distribution in the borehole.

The load calculations should be as per Finite Element Method for the selection of the channels for suitable size of the duct and should be provided by the contractor to the consultant for verification.

Supporting DETA (European Technical Approval) it's for low pressure systems			
Larger Side of Duct mm	Support Channel mm	Vertical Rod Φ mm	Maximum Spacing between supports mm
0-600	27x18x1.2	M8	2400
601-1250	38x24x2	M8	2400
1250-2100	38x40x2	M10	2400
2100 and above	40x60x3	M12	2400

Fig G. Typical Arrangement for Duct, CHW pipe and drain pipe support from building shaft



6.11 INSULATION

The scope of this section comprises the supply and application of insulation conforming to these specifications.

Duct Thermal Insulation:

- Insulation Materials Technical Specifications - Class O Plus Nitrile Rubber.
- Recommended Adhesive
- Summary of Duct Insulation Thickness

- **Insulation Materials Technical Specifications - Class O Plus Nitrile Rubber with Company Bonded Cloth in one**

layer

- Thermal conductivity of elastomeric nitrile rubber shall not exceed 0.033W/ (m*K) at an average temperature of 0º C
- The insulation shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame, also pass Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category requirement as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990
- Material shall be FM (Factory Mutual), USA approved.
- Unlike other generic materials it should be self-extinguishing, does not drip fireballs and does not spread flames
- Moisture Diffusion Resistance Factor or 'μ' value shall be minimum 7,000 for the insulation material without any outer covering or vapor barrier.
- *It shall have antimicrobial product protection*, EPA approved, Antimicrobial Behavior: No fungal growth tested as per DIN EN ISO 846 Method A Standard for (Fungi), Bacterial Resistance: No bacterial growth and 99.9% reduction tested as per DIN EN ISO 846 Method C for (Bacteria),
- Temp Range: Min + 7ºC to Max 105ºC
- Approved by the EPA (ENVIRONMENTAL PROTECTION AGENCY) for use in Ductwork.

1.0 A. **<u>Duct Thermal Insulation (Class O Plus Nitrile Rubber - inside Air-conditioning Areas):</u>**

- Insulation Materials Technical Specifications - **Closed Cell Nitrile Rubber with tough Aluminum foil facing with μ value of 60,000**
- Recommended Adhesive by manufacturer of Nitrile rubber
- Summary of Duct Insulation Thickness

<u>Insulation Materials Technical Specifications - Closed Cell Nitrile Rubber with tough Aluminum Foil facing</u>

- Thermal conductivity of elastomeric nitrile rubber shall not exceed 0.035W/ (m*K) at an average temperature of 0º C (Tested acc. To EN 12667 & EN ISO 8497)
- The insulation material shall be closed cell Elastomeric Nitrile rubber with tough aluminum foil laminated on one side having thickness of 60 microns
- The insulation shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame, also pass Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category requirement as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990
- Insulation should be approved by Local State's Fire Department
- Density of laminated insulation material shall be between 50 to 70 Kg/m³
- Density of foam insulation shall be between 40 to 55 Kg/m³
- The Aluminum foil (Embossed) shall be of 60 microns thickness and weight shall be 193 gsm as per EN ISO 2286-2.
- The material shall be Duct and Fiber Free
- **Unlike other generic materials it should be self-extinguishing, does not drip fireballs and does not spread flames**
- Moisture Diffusion Resistance Factor or **'μ' value shall be minimum 60,000** for an insulation material with lamination or Vapor barrier as per EN 12086 and Min 7000 for base foam without any lamination or vapor barrier to prevent water vapor ingress
- The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero
- The Insulation material shall withstand maximum Surface Temperature of +85 deg C and minimum surface temperature of 0 deg C as per EN 14706
- **Material Color: Black**
- ***Approved by the EPA for use in Duct Work.***

<u>Summary of Duct Insulation Thickness</u>

	Insulation Thickness(mm)	**Material to be used**

Supply Air Duct	19	Class O Closed Cell Nitrile Rubber with tough Aluminum Foil facing
Return Air Duct	13	Class O Closed Cell Nitrile Rubber with tough Aluminum Foil facing

Duct Thermal Insulation: (Exposed to Atmosphere)

- Insulation material technical specification – Class 'O' Nitrile Rubber with company bonded Glass Cloth and above it 2 layers of UV coat shall be applied.
- Recommended adhesive by manufacturer of Nitrile rubber
- Summary of Duct Insulation thickness

Insulation Material Technical Specification – Class O Nitrile Rubber GC Bonded and above it 2 layers of UV paint

- Insulation material shall be **Closed Cell Elastomeric Nitrile Rubber**
- Thermal conductivity of elastomeric nitrile rubber shall not exceed **0.035 W/(m.K**) at mean temperature of 20 deg C. (Tested acc. To EN 12667/ EN ISO 8497)
- The insulation shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame, also pass Fire Propagation requirement as per BS476 Part 6 to meet the **Class 'O'** Fire category requirement as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990
- Material shall be **FM (Factory Mutual), USA approved.**
- Unlike other generic materials it should be self-extinguishing, does not drip and does not spread flames
- Moisture Diffusion Resistance Factor or 'μ' value shall be **minimum 7,000 for the insulation material without any outer covering**

Specifications for Manufacturer Bonded Treated Woven Glass Fiber for Outer covering of Insulation Material

- Temperature Range: 0°C to +105°C Overall (irrespective of the base product) Color: Black
- Treatment: Shall be treated Water Based Acrylic binder to give crisp and non-piling property to the fabric, to help in easy installation, minimize fiber erosion, good aesthetics and resistance to abrasion. Fiber spillage / Thread raveling should be minimum.

Density: 200 +/- 20 gsm

Tensile Strength: 275 +/- 25 Kg / 50 mm (minimum)

Thickness: 0.18 mm / 7 mill

Type: Solvent based Acrylic Adhesive

Peel Strength: 1000 gm / 25 mm (minimum) - (Adhesive to steel)

Summary of Duct Insulation Thickness

	Insulation	Material to be used

	Thickness(mm)	
Supply Air Duct	25	25 mm thick Closed cell Class O Nitrile rubber GC bonded and above it 2 layers of UV paint
Return Air Duct	19	19 mm thick Closed cell Class O Nitrile rubber GC bonded and above it 2 layers of UV paint

Duct Thermal Insulation (Class O Plus EPDM - inside Air-conditioning Areas):

MATERIAL

Thermal insulation material for Duct & Pipe insulation shall be **closed cell Elastomeric EPDM Rubber.** The Thermal conductivity of the insulation material shall not exceed 0.038 W/m°K at an average temperature of 32°C. Density of the EPDM rubber shall be 40-60 Kg/m^3. The product shall have temperature range of –57 °C to 125°C. The insulation material shall be fire rated for Class V 0 as per UL 94. The flammability and smoke density shall be 25/50 as per ASTM E 84, Non-flammable as per JIS K 6911, Australian standard 1530 and class 5.3 as per EMPA. Water vapor diffusion resistance factor (μ) ≥ 7000.The water absorption (weight %) shall not exceed 5 as per ASTM D 1056. The insulation material should be free from Nitrosamine contents as per US FDA norms. It should be CFC free. It should not be corrosive to copper and stainless when tested as per DIN 1988.The material should not develop crack when tested for ozone resistance as per ASTM 1149. The % shrinkage (Heat Stability) should not exceed 6 when tested as per ASTM C 534(93°C, 7 days). No cracks should develop when exposed to UV (accelerated weathering resistance test cycle UVB-313 at 60 °C/8h, CON at 50°C /4h) as per ASTM G 154-04. The resistance to microbiological growth should be in accordance to UL 181 – and meet the acceptance criteria of resistance to fungal contamination as per ASTM G21. It should meet the acceptance criteria of resistance to bacterial contamination as per ASTM 2180.

Each lot of insulation material delivered at site shall be accompanied with manufacturer's test certificate for thermal conductivity values, density, water vapor resistance factor, Nitrosamine content, Heat stability and fire properties. Samples of insulation material from each lot delivered at site may be selected by Owner's site representative and gotten tested for thermal conductivity and density at Contractor's cost. Adhesive used for sealing the insulation shall be modified neoprene contact adhesive with minimum bond strength of 1.25 kg/m2. It should have high water vapor resistance, good weathering properties and self-extinguishing once dried strictly as per manufacturer's recommendations.

Ducting insulation thickness shall be as per table below.

Ducting position	Thickness
SA duct in RA path	19mm
Ducted return air system	SA duct: 25mm RA duct: 19mm
Both SA& RA exposed	Both 25mm

3). Duct Acoustic Insulation:

Open Cell Nitrile Rubber Material with Microban for Antifungal and Antimicrobial Properties:

- Acoustic Material Properties
- The Random Incidence Sound Absorption Coefficient (RISAC); tested as per ISO 354, shall be minimum as per enclosed chart
- Recommended Thickness for Ducts
- Recommended Adhesive by manufacturer of Nitrile rubber

Acoustic Material Properties

- *Acoustic Material should be open cell nitrile rubber.*
- *Its material shall be fiber free.*
- *The density of the same shall be within* ***140-180 Kg/m3****.*
- *It shall have Microban®*; antimicrobial product protection,* Antimicrobial Behavior: No fungal growth tested as per **DIN EN ISO 846** Method A Standard for (Fungi), Bacterial Resistance: **No bacterial growth** and 99.9% reduction tested as per DIN EN ISO 846 Method C for (Bacteria).
- *The material shall have a* ***thermal conductivity*** *not exceeding* ***0.047 W/m.K @ 20 Deg. C***

- *The material shall withstand maximum surface temperature of* ***+85⁰ C*** *and minimum surface temperature of* ***-20 Deg.C.*** *The material shall conform to Class 1 rating for surface spread of Flame in accordance to* ***BS 476 Part 7*** *&* ***UL 94*** *(HBF, HF 1 & HF 2) in accordance to UL 94, 1996.*
- The insulation shall pass ***Air Erosion Resistance Test*** in accordance to ASTM Standard C 1071-05 (section 12.7), from an independent accredited testing agency.
- *Thickness of the material shall be as specified for the individual application.*
- *The insulation shall be installed as per manufacturer's recommendation.*
- ***Approved by the EPA for use in Duct Work***

The Random Incidence Sound Absorption Coefficient (RISAC); tested as per ISO 354, shall be minimum as per enclosed Table No. (1)

Freq(Hz)	**125**	**250**	**500**	**1000**	**2000**	**4000**
10 mm	0	0.05	0.15	0.4	0.85	1
15 mm	0	0.1	0.3	0.75	1	0.9
20 mm	0.05	0.15	0.4	0.9	1	0.95
25 mm	0.05	0.25	0.85	1	0.9	0.95
30 mm	0.1	0.35	0.95	1	0.95	1
40 mm	0.15	0.5	1	0.9	0.95	0.95
50 mm	0.25	0.75	1	1	1	1

Recommended Thickness for Ducts

For Duct sizing lesser than 600 mm X 600 mm: **15 mm thick**

Insulation Thickness will be 15 mm for Duct Acoustic Insulation

4). AHU Room Wall Acoustic Insulation:

- Insulation Material Technical Specifications
- Application Procedure

Product Technical Specifications: Resin Bonded Slab with **SINGLE SIDE** company Bonded BGT (Black Glass Tissue)

Material: Resin Bonded Slab
Density: 140 Kg/M³
Insulation Thickness: 50 mm
Facing: Black Glass Tissue (BGT)

- Thermal Conductivity of Resin Bonded slab shall be 0.037 W/m.K at an average mean temperature of 35 deg C **(as per ASTM C 518)**
- **It should have NRC (Noise Reduction Co-efficient) value of 0.90** at one third octave frequencies – with 25 mm Air gap as per IS: 8225 - 1987, "Measurement of Sound Absorption Coefficient in Reverberation Room" Equivalent to ISO: 354-2003 & ASTM 423-90.

Frequency (Hz)	Sound Absorption Coefficient (α)
500	1.00
1000	1.00
2000	1.00
4000	1.00

- The insulation material (Resin Bonded slab) shall have **Class A** Fire Performance Grade
- Insulation material confirms to **NFPA 90A, 90B** and **LIFE SAFETY CODE NFPA101**
- Insulation material confirms to **ASTM E84** for surface burning characteristics of building materials

- Resin Bonded Rockwool Slabs are manufactured especially for an acoustical application Size: **1000 mm x 500 mm**
- Confirming Standards: IS 8183 and ASTM C612
- The insulation material should be Incombustible (AS PER IS:8183/IS:3144) and moisture resistant & also it should not cause or accelerate any type of corrosion

Application Procedure:

- Clean the surface to be insulated
- Providing & Fixing Resin bonded insulation slabs having 50 mm thickness and Density of 140 Kg/m³ directly on the surface. No framework required to fix it.
- And it's joints shall be sealed with 50 mm wide self-adhesive BGT tape

Insulation Thickness will be 50 mm for AHU Room Thermal Insulation

5). Underdeck Insulation:

Underdeck Insulation:

- Insulation material technical specifications
- Application procedure

Product Technical Specifications: Resin Bonded slab with one Side Company Bonded Aluminum foil
Material: Resin Bonded slab
Density: 140 Kg/M³
Insulation Thickness: 50 mm
Facing: Aluminum foil faced (FSK)

- Thermal Conductivity of Resin Bonded slab shall be ≤ 0.0368 W/m.K at an average mean temperature of 35 deg C **(As per ISO 8301)**
- Insulation material confirms to **NFPA 90A, 90B** and life safety code **NFPA101**
- Insulation material confirms to **ASTM E84** for surface burning characteristics of building materials
- It shall have **R Value of 7.71 hr*ft²*˚F/BTU** (Thermal Resistance Value) – calculated for mean temperature of 35 deg C
- Temperature Range: up to + 750 deg C
- The insulation material (Resin Bonded slab) shall have **Class A** Fire Performance Grade
- Resin Bonded Rockwool Slabs are manufactured for Thermal and acoustical application
- Size: **1000 mm x 500 mm**
- Confirming Standards: IS 8183 and ASTM C612
- The insulation material should be Incombustible and moisture resistant & also it should not cause or accelerate any type of corrosions

Application Procedure:

- Clean the surface to be insulated
- Providing & Fixing Resin Bonded Insulation slabs having density of 140 Kg/m³ and thickness of 50 mm covered with company bonded Aluminum foil below the slab with the help of screw & washer – Cross wire system.
- And joints shall be sealed with 72 mm wide self-adhesive aluminum tape having high strength (70 microns)

Insulation Thickness will be 50 mm for Under deck Insulation

6) **Chilled Water Piping Insulation**:

- Insulation Materials Technical Specifications
- Summary of Chilled Water Piping Insulation Thickness
- **Insulation Material Technical Specification – Class O Nitrile GC Bonded*:***

- Insulation material shall be **Closed Cell Elastomeric Nitrile Rubber**
- Thermal conductivity of elastomeric nitrile rubber shall not exceed **0.035 W/ (m.K**) at an average temperature of 0 C

- The insulation shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame, also pass Fire Propagation requirement as per BS476 Part 6 to meet the **Class 'O'** Fire category requirement as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.

- Unlike other generic materials it should be self-extinguishing, does not drip and does not spread flames

- Moisture Diffusion Resistance Factor or 'μ' value shall be **minimum 7,000 for the insulation material without any out covering**

❖ **Recommended Adhesive (Armaflex Pipe Seal)**

- The adhesive shall be specially formulated for the pipe insulation application and supplied by insulation manufacturer.
- The adhesive shall be Solvent based rubber insulation adhesive, free from benzene.
- It should have excellent Bonding to Porous and Non-Porous Surfaces.
- The adhesive shall be quick drying characteristics for tropical conditions and shall have service temperature of -25ºC to +90ºC.

- **Properties**
- Viscosity @ 30º C :3500 - 4000 cps
- Specific Gravity :0.89 - 0.92 g/cc
- Density : 0.083 kg/m3+ 0.01
- Flash Point : > 10 °C
- Coverage : 4-5 meters of bonded material/liter
- Drying Time : 7 - 8 Minutes
- Tack Retention Time : 25 Minutes

PROTECTIVE COATING OVER INSULATION

❖ **Specifications for Manufacturer Bonded Treated Woven Glass Fiber for Outer covering of Insulation Material**

To provide mechanical strength and protection from damage including UV rays and ozone protection - all pipe / duct insulated with rubber shall be covered with self-adhesive 0.6mm thick rubber protective sheet/tape. The tensile strength of the protection tape should be more than 2.5N/mm as per JIS K6301. The percentage elongation should be more than 50% as per JIS K 6301(speed 50mm/min). The adhesion peel strength should be more than 0.4 kg/25mm as per ASTM D 3330. The initial tack should be less than 1.5 cm as per ASTM D3121. The adhesive shall be strictly as recommended by the manufacturer.

- Temperature Range: 0°C to +105°C Overall (irrespective of the base product) Color: Black
- Treatment: Shall be treated Water Based Acrylic binder to give crisp and non-piling property to the fabric, to help in easy installation, minimize fiber erosion, good aesthetics and resistance to abrasion.
- Fiber spillage / Thread raveling should be minimum.

Density	: 200 +/- 20 gsm
Tensile Strength	: 275 +/- 25 Kg / 50 mm (minimum)
Thickness	: 0.18 mm / 7 mill
Type	: Solvent based Acrylic Adhesive
Peel Strength	: 1000 gm / 25 mm (minimum) - (Adhesive to steel)

❖ **Summary of Pipe Thermal Insulation Thickness Chart & Application:**

Inner Pipe Dia. (NB)	**Required Thickness**	**1st Layer thick(mm)**	**2nd Layer thick(mm)**	**Class 'O' Nitrile Rubber (Self-Sleeve)**
20	25	25	N/A	Self-Sleeve Class 'O' Nitrile with Company Bonded Cloth in one layer
25	32	32	N/A	Self-Sleeve Class 'O' Nitrile with Company Bonded Cloth in one layer
32	32	32	N/A	Self-Sleeve Class 'O' Nitrile with Company Bonded Cloth in one layer
40	32	32	N/A	1st layer Self-Sleeve tubing & 2nd layer manufacturer bonded Treated Woven Glass fiber covering.
50	38	32	6	1st layer Self-Sleeve tubing & 2nd layer manufacturer bonded Treated Woven Glass fiber covering.
65	38	32	6	1st layer Self-Sleeve tubing & 2nd layer manufacturer bonded Treated Woven Glass fiber covering.
80	38	32	6	1st layer Self-Sleeve tubing & 2nd layer

				manufacturer bonded Treated Woven Glass fiber covering.
100	38	32	6	1st layer Self-Sleeve tubing & 2nd layer manufacturer bonded Treated Woven Glass fiber covering.
125	38	32	6	1st layer Self-Sleeve tubing & 2nd layer manufacturer bonded Treated Woven Glass fiber covering.
150	44	19	25	1st & 2nd layer manufacturer Pre-Cut (ready to use) Sheet & 3rd layer of readymade manufacturer's supplied Treated Woven Glass fiber covering (as a separate layer).
200	44	19	25	1st & 2nd layer manufacturer Pre-Cut (ready to use) Sheet & 3rd layer of readymade manufacturer's supplied Treated Woven Glass fiber covering (as a separate layer).
250	44	19	25	1st & 2nd layer manufacturer Pre-Cut (ready to use) Sheet & 3rd layer of readymade manufacturer's supplied Treated Woven Glass fiber covering (as a separate layer).
300	50	25	25	1st & 2nd layer manufacturer Pre-Cut (ready to use) Sheet & 3rd layer of readymade manufacturer's supplied Treated Woven Glass fiber covering (as a separate layer).
350	50	25	25	1st & 2nd layer manufacturer Pre-Cut (ready to use) Sheet & 3rd layer of readymade manufacturer's supplied Treated Woven Glass fiber covering (as a separate layer).
400	50	25	25	1st & 2nd layer manufacturer Pre-Cut (ready to use) Sheet & 3rd layer of readymade manufacturer's supplied Treated Woven Glass fiber covering (as a separate layer).
500	50	25	25	1st & 2nd layer manufacturer Pre-Cut (ready to use) Sheet & 3rd layer of readymade manufacturer's supplied Treated Woven Glass fiber covering (as a separate layer).
600	50	25	25	1st & 2nd layer manufacturer Pre-Cut (ready to use) Sheet & 3rd layer of readymade manufacturer's supplied Treated Woven Glass fiber covering (as a separate layer).
700	50	25	25	1st & 2nd layer manufacturer Pre-Cut (ready to use) Sheet & 3rd layer of readymade manufacturer's supplied Treated Woven Glass fiber covering (as a separate layer).

Recommended Method of Application

1. Clean the substrates to be bonded, so that they are free from Dust/Rust/Oil/Moisture.
2. Stir the Fevicol AC Duct King Lag Coating AF 5590 well before use to have a homogeneous material prior to application.
3. Apply the Fevicol AC Duct King Lag Coating AF 5590 on GI/Aluminum or EPS surface uniformly by brush.
4. Immediately wrap Glass or Canvas cloth tightly with overlapping so that no gaps are left, and no air pockets are formed.
5. Allow to dry and apply additional coating by similar method if required, to achieve smooth neat coated surface.
6. Allow to cure for minimum 24 hrs. at ambient temperature.

Specifications for Protective Coating for Ducting / Chilled Water Piping (Concealed / Exposed Application Areas)

1.1 Protective Coating shall be a flexible, fire resistive fungicidal resistant compound suitable for vapor sealing insulated ducts and pipes.
1.2 Protective Coating shall be suitable for indoor / outdoor use and sustain in high humidity environments.
1.3 Water vapor presence shall not exceed 0.932 metric perms at 1.3 mm dry film thickness when tested in compliance with ASTM E 96.
1.4 When tested for surface burning characteristics in compliance with ASTM E-84, flame spread shall be 5 and smoke developed 10 and complies with ASTM 4804 for flammability resistance.
1.5 Protective Coating shall be UL classified and should conform to UL 723 for surface burning characteristics.
1.6 The Protective Coating should conform to ASTM D 5590 standard for fungal resistance.
1.7 Protective Coating shall be available in color matching the design or façade requirement and UV resistant.
1.8 The Protective Coating should be easy to apply with brush.
1.9 Third Party Test Certification must be furnished by the Manufacturer to substantiate the claims.

7) Drainpipe insulation (CLASS '0' NITRILE - 6 MM THICK SLEEVE WITHOUT BONDED CLOTH & NEITHER SELF SLEEVE NITRILE SHALL BE USED FOR DRAIN PIPING) i.e. only plain nitrile rubber sleeve to be applied with the help of specified adhesive:

a) Insulation Material Technical Specification

b) Recommended Adhesive

(a) <u>Insulation Material Technical Specification</u>

- Insulation material shall be Closed Cell Elastomeric Nitrile Rubber
- Thermal conductivity of elastomeric nitrile rubber shall not exceed 0.035 W/ (m*K) at an average temperature of 0° C
- The insulation shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame, also pass Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category requirement as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990
- **Unlike other generic materials it should be self-extinguishing, does not drip fireballs and does not spread flames**
- Moisture Diffusion Resistance Factor or 'μ' value shall be minimum 7,000 for the insulation material without any out covering or vapor barrier
- Material should be highly flexible so that it can be applied over surfaces like pipes easily. Also, it should be suitable for insulation over non uniform and asymmetrical shapes like valves, flanges etc.

(b) <u>Recommended Adhesive</u>

- The adhesive shall be specially formulated for the pipe insulation application and supplied by insulation manufacturer.
- The adhesive shall be Solvent based rubber insulation adhesive, free from benzene.
- It should have excellent Bonding to Porous and Non-Porous Surfaces.
- The adhesive shall be quick drying characteristics for tropical conditions and shall have service temperature of -25ºC to +90ºC.

- **<u>Properties</u>**

• Viscosity @ 30º C : 3500 - 4000 cps
• Specific Gravity : 0.89 - 0.92 g/cc
* Density : 0.083 kg/m3+ 0.01
* Flash Point : > 10 °C
* Coverage : 4-5 meters of bonded material/liter
* Drying Time : 7 - 8 Minutes
* Tack Retention Time : 25 Minutes

Special Remark: PRODUCT SPECIFICATIONS:

- 06 MM THICK NITRILE SLEEVE FOR DRAIN PIPING INSULATION.
- IT SHALL BE PLAIN SLEEVE WITHOUT BONDED & WITHOUT SELF SLEEVE.

An additional layer of 19 mm thick Class 'O' Nitrile must be applying for chilled water piping installed in low ventilation areas (e.g. shafts, tunnel etc.)

9. <u>False Ceiling Insulation (Thermo-Acoustic)</u>

- Insulation Material Technical Specifications
- Application Procedure

- **<u>Product Technical Specifications:</u>** Resin Bonded Slab with single side company bonded Black Glass Tissue (BGT)

<u>Material:</u> Resin Bonded Slab
<u>Density:</u> 140 Kg/M³
<u>Insulation Thickness:</u> 50 mm
<u>Facing:</u> Black Glass Tissue (BGT)

- Thermal Conductivity of Resin Bonded slab shall be 0.037 W/m.K at an average mean temperature of **35 deg C** (As per ASTM C 518)
- Insulation material confirms to **NFPA 90A, 90B** and **LIFE SAFETY CODE NFPA101**
- Insulation material confirms to **ASTM E84** for surface burning characteristics of building materials
- It shall have **R Value of 7.67 hr*ft²*˚F/BTU** (Thermal Resistance Value) – calculated for mean temperature of 35 deg C

- **<u>It should have NRC (Noise Reduction Co-efficient) value of 0.90</u>** at one third octave frequencies – with 25 mm Air gap as per IS: 8225 - 1987, "Measurement of Sound Absorption Coefficient in Reverberation Room" Equivalent to ISO: 354-2003 & ASTM 423-90

Frequency (Hz)	Sound Absorption Coefficient (α)
500	1.00
1000	1.00
2000	1.00
4000	1.00

- The insulation material (Resin Bonded slab) shall have **Class A** Fire Performance Grade
- Resin Bonded Rockwool Slabs are manufactured especially for an acoustical application
- Size: **<u>1000 mm x 500 mm</u>**
- Confirming Standards: IS 8183 and ASTM C612
- The insulation material should be Incombustible **(AS PER IS:8183/IS:3144)** and moisture resistant & also it should not cause or accelerate any type of corrosion

- **Insulation Thickness will be 50 mm for False Ceiling Insulation**

METHOD OF MEASUREMENT FOR INSULATION:

Measurement of duct, piping, equipment, plant room and false ceiling insulation shall be taken as per following basis.

All duct thermal insulation and acoustic duct lining shall be measured on the basis of prime surface area (linear measured bare duct surface area), excluding all openings for grilles and diffusers and including all flanges, dampers etc. Thus, the surface area for thermally insulated duct as well as acoustically lined duct shall be equal to perimeter (comprising outer width and depth of bare duct to be insulated) multiplied by centerline length of duct including all tapered pieces, tees, bends, branches etc.

Support structure required for the ducting support will not be considered as an extra. Acoustic insulation of AHU and A.C. plant room shall be measured on the basis of finished surface area. Any method if specified in under Special Purchase conditions other than above shall overrule.

6.12 AIR DISTRIBUTION EQUIPMENT

DAMPERS/ FIRE DAMPER

All dampers shall be of 18 S.W.G. G.I sheets louver dampers of robust construction and tight fitting. The design, method of handling and control, shall be suitable for the location and service required. Dampers shall be provided with suitable links, levers and quadrants as required for their proper operation, control or setting in any desired position. Dampers and their operating devices shall be made robust, easily operable and accessible through suitable access door. Every damper shall always have indication device clearly showing the damper position. All the bushing will be of brass only.

A fire damper shall be provided between each air handling unit room and the rest area and at crossing of fire rated walls. The fire dampers shall be confirming to UL-655 and other applicable fire codes. The dampers shall be operated through either fusible link or solenoid valve.

FIRE DAMPER – (FUSIBLE LINK OPERATED)

Fire Dampers shall be installed in the openings / passage formed by Supply and Return air ducts when they pass through AHU room walls as well as when passing through all floors as shown in the drawing.

Fire Dampers shall be operated with at least 90 minutes' fire rating as per UL 555-1995 as tested by CBRI Roorkee, India.

Fire Dampers shall be multi leaf type. The blades & outer frame shall be made of 16G galvanized steel sheet. Fire damper assembly shall be factory fitted in a sleeve made of 18 G galvanized steel sheet of minimum 400 mm long. The blades shall be pivoted on both sides using chrome-plated spindles in self-lubricated bronze bushes. Metallic